Appendix B

Basic probability theory

Includes a description of statistical distance.

B.1 Birthday Paradox

Theorem B.1. *Let $\mathcal{M}$ be a set of size n and let $X_1, \ldots, X_k$ be k independent random variables uniform in $\mathcal{M}$. Let C be the event that for some distinct $i, j \in \{1, \ldots, k\}$ we have that $X_i = X_j$. Then*

(i) $\quad \Pr[C] \geq 1 - e^{-k(k-1)/2n} \geq \min\left\{\frac{k(k-1)}{4n}, 0.63\right\}$, *and*

(ii) $\quad \Pr[C] \leq 1 - e^{-k(k-1)/n}$ *when $k < n/2$.*

Proof. These all follow easily from the inequality

$$1 - x \leq e^{-x} \leq 1 - x/2,$$

which holds for all $x \in [0, 1]$. □

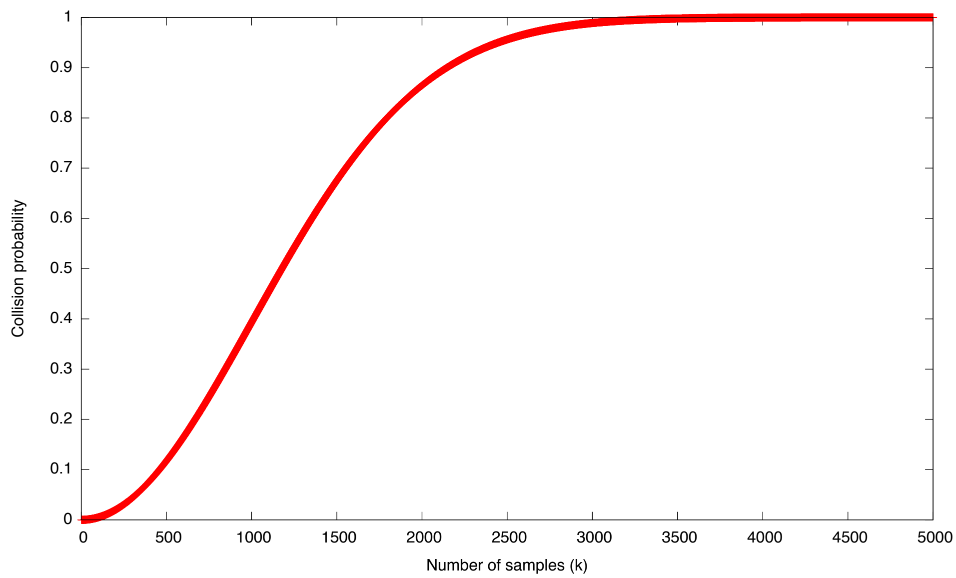
Most frequently we will use the lower bound to say that a collision happens with *at least* a certain probability. But occasionally we will use the upper bound to argue that collisions do not happen.

It is well documented that birthdays are not really uniform throughout the year. For example, in the U.S. the percentage of births in September is higher than in any other month. We show next that this non-uniformity only increases the probability of collision.

We present a stronger version of the birthday paradox that applies to independent random variables that are not necessarily uniform in $\mathcal{M}$. We do, however, require that all random variables are identically distributed. Such random variables are called i.i.d (independent and identically distributed). This version of the birthday paradox is due to Blom [Blom, D. (1973), "A birthday problem", American Mathematical Monthly, vol. 80, pp. 1141-1142].

Corollary B.2. *Let $\mathcal{M}$ be a set of size n and let $X_1, \ldots, X_k$ be k i.i.d random variables over $\mathcal{M}$ where $k \geq 2$. Let C be the event that for some distinct $i, j \in \{1, \ldots, k\}$ we have that $X_i = X_j$. Then*

$$\Pr[C] \geq 1 - e^{-k(k-1)/2n} \geq \min\left\{\frac{k(k-1)}{4n}, 0.63\right\}.$$



The graph shows that collision probability for $n = 10^6$ elements and k ranging from one sample to 5000 samples. It illustrates the threshold phenomenon around the square root. At the square root, $\sqrt{n} = 1000$, the collision probability is about 0.4. Already at $4\sqrt{n} = 4000$ the collision probability is almost 1. At $0.5\sqrt{n} = 500$ the collision probability is small.

Figure B.1: Birthday Paradox

Proof. Let X be a random variable distributed as X_1. Let $\mathcal{M} = \{a_1, \ldots, a_n\}$ and let $p_i = \Pr[X = a_i]$. Let I be the set of all k-tuples over $\mathcal{M}$ containing distinct elements. Then I contains $\binom{n}{k}k!$ tuples. Since the variables are independent we have that:

$$\Pr[\neg C] = \sum_{(b_1,\ldots,b_k)\in I} \Pr[X_1 = b_1 \wedge \ldots \wedge X_k = b_k] = \sum_{(b_1,\ldots,b_k)\in I} \prod_{j=1}^{k} p_{b_j} \tag{B.1}$$

We show that this sum is maximized when $p_1 = p_2 = \ldots = p_n = 1/n$. This will mean that the probability of collision is minimized when all the variables are uniform. The Corollary will then follow from Theorem B.1.

Suppose some p_i is not $1/n$, say $p_i < 1/n$. Since $\sum_{j=1}^{n} p_i = 1$ there must be another p_j such that $p_j > 1/n$. Let $\epsilon = \min((1/n) - p_i,\ p_j - 1/n)$ and note that $p_j - p_i > \epsilon$. We show that replacing p_i by $p_i + \epsilon$ and p_j by $p_j - \epsilon$ increases the value of the sum in (B.1). Clearly, the resulting $p_1, \ldots, p_n$ still sum to 1. Hence, the resulting $p_1, \ldots, p_n$ form a distribution over $\mathcal{M}$ in which there is one less value that is not $1/n$. Furthermore, the probability of no collision in this distribution is greater than in the unmodified distribution. Repeating this replacement process at most n times will show that the sum is maximized when all the p_i's are equal to $1/n$. Again, this means that the probability of not getting a collision is maximized when the variables are uniform.

Now, consider the sum in (B.1). There are four types of terms. First, there are terms that do not contain either p_i or p_j. These terms are unaffected by the change to p_i and p_j. Second, there are terms that contain exactly one of p_i or p_j. These terms pair up. For every k-tuple that contains i but not j there is a corresponding tuple that contains j but not i. Then the sum of the corresponding two terms in (B.1) looks like $A(p_i + \epsilon) + A(p_j - \epsilon)$ for some $A \in [0, 1]$. Since this equals $Ap_i + Ap_j$ the sum of these two terms is not affected by the change to p_i and p_j. Finally, there are terms in (B.1) that contain both p_i and p_j. These terms change by

$$B(p_i + \epsilon)(p_j - \epsilon) - Bp_ip_j = B[\epsilon(p_j - p_i) - \epsilon^2]$$

for some $B \in [0, 1]$. By definition of ϵ we know that $p_j - p_i > \epsilon$ and therefore $\epsilon(p_j - p_i) - \epsilon^2 > 0$. Hence, the sum with modified p_i and p_j is larger than the sum with the unmodified values.

Overall, we proved that the modification to p_i and p_j increases the value of the sum in (B.1), as required. This completes the proof of the Corollary. $\square$

B.1.1 More collision bounds

Consider the sequence $x_i \leftarrow f(x_{i-1})$ for a random function $f : \mathcal{X} \to \mathcal{X}$. Analyze the cycle time of this walk (needed for Pollard). Now, consider the same sequence for a permutation $\pi : \mathcal{X} \to \mathcal{X}$. Analyze the cycle time (needed for analysis of SecurID identification).

B.1.2 A simple distinguisher

We describe a simple algorithm that distinguishes two distributions on strings in $\{0, 1\}^n$. Let $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$ be independent random variables taking values in $\{0, 1\}$. Then

$$X := (X_1, \ldots, X_n) \qquad \text{and} \qquad Y := (Y_1, \ldots, Y_n)$$

are elements of $\{0, 1\}^n$. Suppose, that for $i = 1, \ldots, n$ we have

$$\Pr[X_i = 1] = p \qquad \text{and} \qquad \Pr[Y_i = 1] = (1 + 2\epsilon) \cdot p$$

for some $p \in [0,1]$ and some small $\epsilon > 0$ so that $(1+2\epsilon) \cdot p \leq 1$. Then X and Y induce two distinct distributions on $\{0,1\}^n$.

We are given an n-bit string T and are told that it is either sampled according to the distribution X or the distribution Y, so that both p and ϵ are known to us. Our goal is to decide which distribution T was sampled from. Consider the following simple algorithm $\mathcal{A}$:

> input: $T = (t_1, \ldots, t_n) \in \{0,1\}^n$
> output: 1 if T is sampled from X and 0 otherwise
>
> $s \leftarrow (1/n) \cdot \sum_{i=1}^{n} t_i$
> if $s > p \cdot (1+\epsilon)$ output 0 else output 1

We are primarily interested in the quantity

$$\Delta := \big| \Pr[\mathcal{A}(T_x) = 1] - \Pr[\mathcal{A}(T_y) = 1] \big| \quad \in [0,1]$$

where $T_x \xleftarrow{\text{R}} X$ and $T_y \xleftarrow{\text{R}} Y$. This quantity captures how well $\mathcal{A}$ distinguishes the distributions X and Y. For a good distinguisher Δ will be close to 1. For a weak distinguisher Δ will be close to 0. The following theorem shows that when n is about $1/(p\epsilon^2)$ then Δ is about $1/2$.

Theorem B.3. *For all $p \in [0,1]$ and $\epsilon < 0.3$, if $n = 4\lceil 1/(p\epsilon^2) \rceil$ then $\Delta > 0.5$*

Proof. The proof follows directly from the Chernoff bound. When T is sampled from X the Chernoff bound implies that

$$\Pr[\mathcal{A}(T_x) = 1] = \Pr[s > p(1+\epsilon)] \leq e^{-n \cdot (p\epsilon^2/2)} \leq e^{-2} \leq 0.135$$

When T is sampled from Y then the Chernoff bound implies that

$$\Pr[\mathcal{A}(T_y) = 0] = \Pr[s < p(1+\epsilon)] \leq e^{-n \cdot (p\epsilon^2/4)} \leq e^{-1} \leq 0.368$$

Hence, $\Delta > |(1 - 0.368) - 0.135| = 0.503$ and the bound follows. $\square$